

THE UNIVERSITY OF WARWICK

Paper Details

Paper Code: CS3550

Paper Title: Digital Forensics

Exam Period: Summer 2025

Exam Rubric

Time Allowed: 2 hours

Exam Type: Standard Examination

Approved Calculators: Permitted

Additional Stationery

Instructions

Answer **ALL** questions from Section A and **TWO** questions from Section B.

If more than **TWO** questions are answered from Section B, please clearly indicate which questions you wish to be marked. Otherwise, the first **TWO** questions of this section answered in the answer booklet will be marked.

Read carefully the instructions on the answer booklet and make sure that the particulars required are entered on each answer booklet.

The exam question paper **MUST NOT** be removed from the examination venue.

Permitted calculators: CASIO FX82, FX83 and FX85 series.

Section A Answer **ALL** questions.

Question 1

[25 marks]

- (a) You have converted an RGB image to Y'CbCr color space. To reduce color redundancy, you are about to perform Chroma subsampling on Cb/Cr color channel(s). Given a certain Cb/Cr color channel **I**:

$$\mathbf{I} = \begin{bmatrix} 31 & 26 & 23 & 21 \\ 28 & 24 & 21 & 22 \\ 26 & 23 & 20 & 22 \\ 22 & 20 & 20 & 21 \end{bmatrix},$$

you are planning to subsample it in two different formats: 4 : 2 : 0 (left) and 4 : 1 : 1 (direct) formats. Show your subsampled results (use `floor` function to output integers). [6]

- (b) Please answer the following two questions on spatial domain based image denoising:
- (i) When there is Gaussian noise in an image, a mean filter can be used for image smoothing, and different filter sizes may have different effects. What is the potential consequence if the mean filter size is too large? [4]
- (ii) There is an image with some salt and pepper noise (i.e., dead pixels) caused by

faulty sensors. Here is an example image patch: $\mathbf{J} = \begin{bmatrix} 19 & 22 & 17 \\ 21 & 255 & 16 \\ 20 & 18 & 17 \end{bmatrix}$.

You want to enhance the image quality by using the median filter. Calculate the filtered centre pixel (of **J**) based on mean and median filters respectively, and justify why median filter is a better choice at removing dead pixels. [6]

- (c) Gamma correction and histogram equalization are two popular pixel-domain based contrast enhancement techniques. Describe these techniques, including their major differences. [6]
- (d) Histogram equalization is a popular method for image contrast enhancement. Please explain why contrast can be enhanced by flattening the probability mass function (PMF) of the target image. [3]

Question 2

[25 marks]

- (a) Given an image $\mathbf{I} = \begin{bmatrix} 66 & 11 & 66 & 44 & 11 \\ 11 & 66 & 66 & 66 & 11 \\ 66 & 66 & 22 & 44 & 11 \\ 66 & 66 & 22 & 44 & 66 \end{bmatrix}$, answer the following questions:
- (i) Plot the image's histogram. [3]
 - (ii) In the fixed-length coding scheme, at least how many bits are required to code each entry in $\mathbf{I}$? [3]
 - (iii) The general idea of Huffman coding is to assign short binary codewords to symbols with higher probabilities, and longer codewords to those with lower probabilities. Please list one set of Huffman code for $\mathbf{I}$ (Non-unique answer). [4]
 - (iv) Based on the Huffman codewords in (iii), calculate the average code length. [3]
- (b) JPEG compression technique has been designed to remove three types of redundancy, namely, spatial redundancy, psychovisual redundancy, and coding redundancy. Answer the following questions:
- (i) What is spatial redundancy? [2]
 - (ii) What is the key operation for spatial redundancy removal? [2]
 - (iii) Of the three redundancy removals, which one is a lossy process? [2]
- (c) Given the coefficient block $\mathbf{C}$, and a fixed quantisation factor $q = 5$
- $$\mathbf{C} = \begin{bmatrix} 63 & 33 & 20 & 8 \\ 41 & 30 & 18 & 4 \\ 20 & 25 & 4 & 2 \\ 10 & 15 & 2 & 1 \end{bmatrix}, \text{ please calculate the following:}$$
- (i) Quantized coefficient block $\mathbf{C}'$ (use `round` function during quantization); [2]
 - (ii) De-quantized coefficient block $\mathbf{C}''$; [2]
 - (iii) Quantization error in terms of SSD (sum of squared difference). [2]

Section B Answer **TWO** questions.

Question 3

[25 marks]

- (a) Given a 2×2 image patch $\mathbf{I} = \begin{bmatrix} 33 & 16 \\ 15 & 129 \end{bmatrix}$, answer the following questions:
- (i) Give the binary representation of $\mathbf{I}$; and show its bit-plane 1 and bit-plane 6 [6]
 - (ii) Describe how to create visible and invisible watermarks using bit-plane substitution methods in the pixel domain. [4]
- (b) Watermark embedding can also be performed in the frequency domain, which is considered to be more robust than pixel-domain based approaches. Is it a good idea to embed watermark in high-frequency components? Justify your answer. [6]
- (c) For video compression, video sequences are divided into Group-of-Pictures (GOPs). Answer the following questions:
- (i) What are the three frame types of images within an GOP? [3]
 - (ii) In 100 words or less, briefly explain how to encode these three types of frames within a GOP. [6]
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Question 4

[25 marks]

- (a) It is common practice to perform filtering in the frequency domain for image processing. Answer the following questions:
- (i) In Fig. 1, is the mask $H(u, v)$ a high-pass filter or low-pass filter? Please explain. [3]
- (ii) Based on (i), please describe what is expected of the output image $\hat{f}(x, y)$ [3]

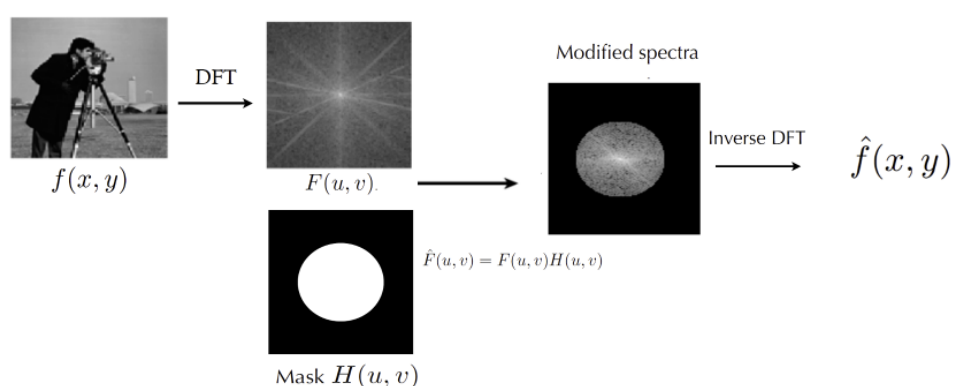


Figure 1: Image filtering in the frequency domain

- (iii) You want to remove the repetitive patterns in Fig. 2 by using filtering techniques in the frequency domain. Please list the key steps. [7]



Figure 2: an image with repetitive patterns

- (b) Circular shift&match is an exhaustive search approach to detect copy-move forgeries. During matching (between the original and the shifted images), there may exist false positives or false negatives. Explain some image processing approaches for removing these two types of errors. [4]

(c) Local Binary Pattern (LBP) is a popular feature template widely used for visual recognition, which is less sensitive to illumination changes. Answer the following two questions:

- (i) Given a 3×3 image patch $\mathbf{J} = \begin{bmatrix} 70 & 22 & 44 \\ 20 & 50 & 80 \\ 90 & 60 & 32 \end{bmatrix}$, please calculate the LBP code of the center pixel, in both binary and decimal forms. [4]
- (ii) Explain why LBP feature is less sensitive to illumination changes. [4]
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Question 5

[25 marks]

- (a) Sensor Pattern Noise (SPN) is a unique signature that can be used for source camera identification. However, it may be affected by image content (e.g., scene details). Please answer the following three questions:
- (i) From the perspective of signal frequency, explain why SPN can be affected by scene details. [2]
 - (ii) Why can't SPN be calculated directly? [3]
 - (iii) Assuming you have several images from a camera, please list the key steps for SPN estimation. [7]
- (b) As generative AI technologies advance, deepfake images may have a profound societal impact, making the development of detection technologies essential. Please answer the following two questions:
- (i) You are designing a deepfake fingerprint visualization system for Generative Adversarial Network (GAN) models. Please provide your design solution and explain how to mitigate the influence of image content in GAN-generated deepfakes. [6]
 - (ii) Machine learning and deep learning approaches are commonly used to train deepfake detectors. Suppose you have trained a detector using deepfakes generated by various GAN models. Discuss the key limitations of this trained detector and propose an improved solution to address these challenges, providing justification for your approach. [7]